



Artificial intelligence-assisted diagnosis of hematologic diseases based on bone marrow smears using deep neural networks

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ABSTRACT

Background and objectives: The morphological examination of bone marrow (BM) cells is essential in both diagnosing and treating various hematologic diseases. However, it is still done manually with a heavy workload. An artificial intelligence-assisted diagnosis support system of BM cells is highly required to reduce the workloads of examiners and improve the reproducibility of the results.

Methods: In this paper, we proposed an artificial intelligence-assisted diagnosis support system of morphological examination based on bone marrow smears including cells detection, classification and prediction of leukemia types. For cell detection, we trained the novel YOLOX-s model to locate cells precisely and obtain single cell images. For cell classification, we regarded it as a fine-grained classification task and proposed a novel architecture called MLFL-Net utilizing multi-level features. Furthermore, we predicted the leukemia types on a dataset including 40 normal people (BM transplantation donors) and 40 patients of different kinds of acute leukemia according to the World Health Organization (WHO) standard.

Results: We constructed a large-scale data set of 11,788 fully-annotated micrographs from 728 smears and 131,300 expert-annotated single cell images. With the data set, the detection model achieved 0.9797 AUC and 4.33% box placement error. For cell classification, the total accuracy of our proposed MLFL-Net reached 89.53% which outperformed all the other related models in identifying cell categories. In the meantime, we took acute leukemia as an example to explore the leukemia types prediction procedure of hematological disease. It generated the same diagnostic prediction as the experts gave for 92.5 percent of the cohort.

Conclusion: This Artificial Intelligence-assisted system can be implemented to aid in clinical decision making and accelerate diagnosis. The method will contribute to promote the intelligence and modernization of BM cytomorphology, which has vital significance of the development of the medical career.

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1. Introduction

The morphological examination of bone marrow (BM) cells is the footstone in both diagnosing and treating various hematologic diseases. Although plenty of complicated methods including cytogenetics, immunophenotyping and molecular genetics are available, the morphological examination remains the basic item and the first key step in diagnosing and treatment response monitoring [1]. Examiners first observe the BM smear under a low magnification to determine the best region for cell counting, then they employ an 100x objective lens with the oil immersion to discriminate and count different kinds of cells. According to the key ratios and the abnormal information of cells, examiners obtain the diagnostic opinions of different types of diseases [2]. However, such a manual

procedure is tedious, labor-intensive, time-consuming and highly dependent on examiner skills [3]. Meanwhile, results are found to be subject to substantial inter-and-intra rater variability [4]. Therefore, an artificial intelligence-assisted diagnosis support system for the morphological examination of BM cells is highly required to improve the efficiency of the examination and reduce the workloads of examiners, which can further promote the development of medical science.

At present, many researchers have made attempts on the artificial intelligence-assisted morphological examination based on peripheral blood (PB) smears [5–8]. However, compared with PB smears, cells in BM smears are more diverse in the aspects of categories and morphology. Meanwhile, cells in BM smears are generally facing problems of high density and severe adhesion, which increases the difficulty of research. Therefore, attempts done on BM smears are relatively few.

Among these attempts, many of them adopted traditional methods for cell segmentation and classification with artificial features

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[9–11], which highly depend on manual labor with low robustness. Since convolutional neural network (CNN) has gained success in other aspects in the medical field [12,13], some institutes attempted to utilize it to explore the key steps for the artificial intelligence-assisted morphological examination. For example, Chandradevan et al. [3] adopted Faster R-CNN [14] to detect cells in microscope images and used VGG-16 [15] to classify them into 12 categories. Fu et al. [16] utilized K-means cluster algorithm and decision tree algorithm to segment cells and classified them with a 27-layer CNN. Matek et al. [17] formed a data set of BM cells and used ResNeXt-50 [18] to classify cells into 21 categories, whose dataset was the largest expert-annotated pool of BM single cell images available so far. Guo et al. [19] presented a class balance classification method for classifying 15 types of bone marrow cell datasets with a class imbalance problem. Maurya et al. [20] developed a novel machine learning-based framework for the classification of bioimages across nine diverse datasets. They classified haematological images into 5 categories and achieved 88% accuracy. However, there was still room for improvement of previous attempts. Some researches focused on low numbers of samples or the categories were not detailed enough, which led to a relatively poor versatility. Meanwhile, most models they used for classification are classical CNN architectures without pertinent designs for BM smears, which limited its performance. In addition, the whole artificial intelligence-assisted diagnosis support system have not been currently practiced in its entirety. The majority of previous studies only focused on the part of the whole system.

In this paper, we constructed a large-scale data set of 11,788 fully-annotated micrographs from 728 smears and 131,300 expert-annotated single cell images, which was literally the important foundation of our research. With the data set, we explored the whole procedure of the artificial intelligence-assisted diagnosis support system based on BM smears including cell detection, cell classification and prediction of leukemia types. For cell detection, we trained the novel YOLOX-s model to locate cells precisely and obtain single cell images. For cell classification, unlike other related work, we regarded it as a fine-grained classification task. We proposed a novel architecture called MLFL-Net utilizing multi-level features and achieved a high accuracy. Furthermore, we evaluated the system on a dataset including 40 normal people (BM transplantation donors) and 40 patients of different kinds of acute leukemia according to the World Health Organization (WHO) standard [21]. The results showed the reliability to some extent.

Our study is of great significance for the assisted diagnosis of hematologic diseases. The main contributions of the current manuscript are as follows:

- An artificial intelligence-assisted diagnosis support system of morphological examination based on bone marrow smears was proposed including cells detection, classification and prediction of leukemia types.
- A novel classification model called MLFL-Net was proposed which utilized multi-level features to classify BM cells and got high accuracy for the fine-grained classification.
- A large-scale dataset was constructed which included 11,788 fully-annotated micrographs from 728 smears and 131,300 expert-annotated single cell images.

2. Methods

We explored the whole procedure of the artificial intelligence assisted diagnosis support system based on BM smears including cell detection, cell classification and prediction of leukemia types as shown in Fig. 1. In this section, we present the construction of the dataset, the training strategy of cell detection model, the

framework of proposed MLFL-Net and the process for prediction of leukemia types in turn.

2.1. Construction of the data set

Deep learning algorithms rely on large-scale expert annotated data sets. Therefore, we constructed a large-scale data set of 11,788 fully-annotated micrographs and 131,300 expert-annotated single cell images. The construction process was as follows.

According to the clinical diagnosis, we collected 728 BM smears in Beijing University People's Hospital from January 2019 to December 2020, which were from 362 patients diagnosed with a variety of hematological diseases and 366 BM transplantation donors. The age range of the cohort was from 11 to 89 years old, which was literally representative and met the clinical demands. Significantly, we included BM smears collected for examinations before and had been preserved in the database of the hospital instead of especially recruiting volunteers. All data have been desensitized without any personally identifiable information.

The BM smears were processed by the Wright Stain and digitized with the microscope (Olympus BX50) and the camera (JEDA SmartV 650D). Firstly, the entire smear was captured by experts under a 10x objective lens in the view of the microscope to select the region of interest. Secondly, 10 to 20 micrographs for each smear were captured under a 10x eyepiece and an 100x oil immersion objective. The numerical aperture of oil immersion objective was 1.3 and the dimensions of the micrographs were 1920×2560 . Some examples are shown in Fig. 2A. Finally, by virtue of the labeling software we developed, experts localized cells in the micrographs and determined the ground truth category label for every single cell. For a single smear, we obtained the final diagnostic opinion and 200 to 300 valid cells with bounding boxes in accordance with routine practice.

For the sake of research, we converted the annotations of micrographs into VOC format and cropped single cells according to the annotations. In view of the data set quality, we excluded cells with blurs or incomplete contours. According to the development and evolution of BM cells, we selected 19 fine-grained categories in 5 cell series (including Lymphopoiesis, Plasmacytopoiesis, Granulopoiesis, Monopoiesis, Erythropoiesis) highly correlated to the clinical diagnosis as shown in Fig. 2B. Finally, the dataset included 11,788 micrographs and 131,300 single cell images. The quantity distribution of the single cells was shown in Fig. 2C.

2.2. Model for cell detection

In this study, we followed the procedure of detection first and classification later for better accuracy. During the detection stage, we trained the state-of-the-art YOLOX-s [22] model with high accuracy and efficiency developed by Megvii to locate cells in micrographs and obtain the single cell images. The number of the output nodes was adjusted into 2 (representing 'Cell' and 'Not Cell' respectively) and the threshold was set to be 0.4 for Non-Maximum Suppression (NMS). The model can generate bounding boxes for detected cells in micrographs and output the probability of each box for belonging to a real cell. Finally, we cropped the original micrographs according to the bounding boxes and got segmented single cells as the inputs of cell classification model.

In this part, we divided our micrographs data set into the train and the validation set with the ratio 8:2, among which 9430 micrographs were utilized to train it and the other 2358 micrographs were utilized to evaluate it. The network was trained on PyTorch and it took about 46 hours of computing time in total. Ultimately, the best performed one in the evaluation phase was selected as the final model.

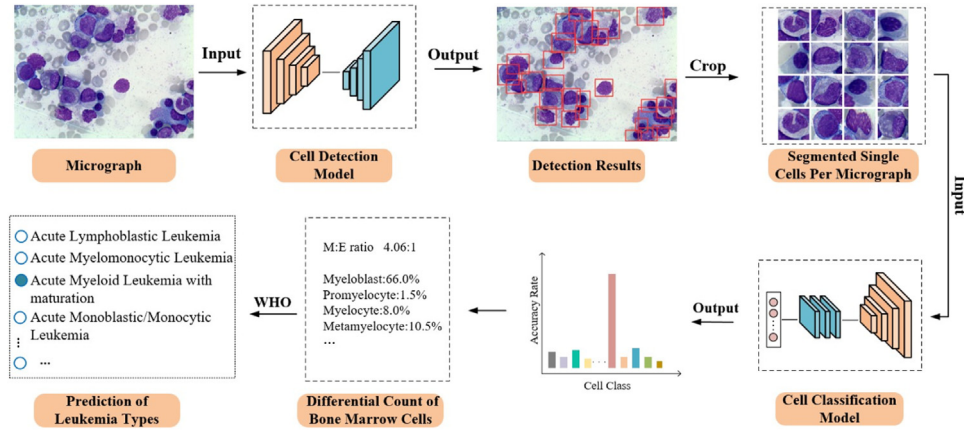


Fig. 1. The artificial intelligence-assisted diagnostic procedure of morphological examination. The procedure includes cell detection, cell classification and prediction of leukemia types. All smears were stained using the Wright Stain, and imaged at $\times 100$ magnification.

2.3. Model for cell classification (MLFL-Net)

After the detection stage, single cell images were obtained. The crucial step in the morphological examination is to classify cells into correct fine-grained categories. As illustrated, for a deep learning network, more semantic information is consisted in high levels and more local detailed information is contained in low levels [23]. In the fine-grained classification task, we need not only the semantic information but also the local detailed one. Motivated by it, we designed a novel model called Multi-Level Feature Learning Network (MLFL-Net) to integrate different kinds of information together by utilizing features at different levels. In the meantime, we enhanced the feature by mining complementary channel information motivated by the self-attention mechanism [24,25]. To our knowledge, the proposed model is the first one to aggregate information with different granularities for the fine-grained classification of BM cells.

As shown in Fig. 3A, MLFL-Net included the backbone model, Multi-Level Feature Learning (MLFL) branch, Fusion Information Distillation (FID) branch and Channel Complementary Information Mining (CCIM) modules. Specifically, we adopted ResNet-50 [26] as our backbone model and divided it into five stages. Instead of using the single feature from the deepest layer as common, we utilized three levels of features respectively from block2, block3 and block4 to gain information with different granularities. To better illustrate our model, we denote these three features as F_{low} , F_{middle} and F_{high} , respectively.

As illustrated, different channels generally correspond to different visual patterns [27] and most of the channels are semantically complementary to the others. Hence, we built CCIM modules to mine complementary information in different channels. In this module, we first calculated the negative correlations between channels and aggregated the complementary information into the original feature as shown in Fig. 3B. Specifically, given the original feature $F_{ori} \in R^{C \times H \times W}$, the correlation matrix of channels $M_{rel} \in R^{C \times C}$ can be calculated as follows:

$$M_{rel} = \text{Softmax}(-F_{flat} F_{flat}^T)$$

where $F_{flat} \in R^{C \times L}$, $L = H \times W$ was reshaped from F_{ori} and F_{flat}^T was transposed from F_{flat} . The Softmax function was adopted in the last dimension and the negative sign here was designed for capturing the negative correlations. After obtaining M_{rel} , we generated a new feature with the complementary information by matrix multiplication. Meanwhile, we aggregated the generated feature with the original one to prevent it from losing important information. The

enhanced feature F_{enh} can be calculated as follows:

$$F_{enh} = F_{ori} + f(M_{rel} \cdot F_{flat})$$

where f here referred to the dimension transformation and a convolutional layer. We sent F_{low} , F_{middle} and F_{high} to CCIM modules to gain three enhanced features and then sent them into the MLFL branch.

The MLFL branch consisted of feature learning modules, the global top-k pooling (GTP) [28] layer and classifiers. The feature learning module as shown in Fig. 3C included two convolutional layers respectively with the kernel size 1 and 3. The former one was utilized to adjust the channel number for different features and the latter one was utilized for detailed information mining. The classifier as shown in Fig. 3D included two fully connected layers, two BN layers, one ReLU layer and a Softmax layer. Meanwhile, we introduced GTP to form feature vectors, which can be regarded as the compromise of global average pooling (GAP) and global max pooling (GMP). Specifically, we calculated the mean of top-k activations in each channel, which was proved to be an effective strategy for fine-grained classification. It alleviated the problem of complex backgrounds compared with GAP and prevented the model from losing important information compared with GMP. In the end, we obtained three probabilities denoted as p_1 , p_2 and p_3 from three classifiers.

In the FID branch, we concatenated three feature vectors in the MLFL branch to gain the fusion feature vector containing information with different granularities. With this feature vector, we trained a fusion classifier to generate a probability denoted as p_{concat} . To introduce the information into different levels, we minimized the Kullback-Leibler (KL) divergence between the classification probability from the fusion classifier and classifiers at different levels. In this way, we minimized the diversity between different classifiers, which can be considered as a kind of knowledge distillation [29].

We divided our single cell image data set into the train and the validation set with the ratio 8:2 using the hierarchical sampling method. Before training the network, we adopted the random cropping method on the input data and resized it into the size of 224×224 . Then we adopted data augmentation methods including random flipping, random rotating and random color jittering. We trained the model with Momentum SGD. All the experiments were conducted on PyTorch.

In this stage, we adopted Cross Entropy (CE) loss and KL loss to form the total loss function. The total loss was as follows:

$$\text{Loss} = \alpha \cdot \left[CE(p_{concat}, y) + \sum_{i=1}^3 CE(p_i, y) \right] + \beta \cdot \sum_{i=1}^3 D_{KL}(p_{concat} || p_i)$$

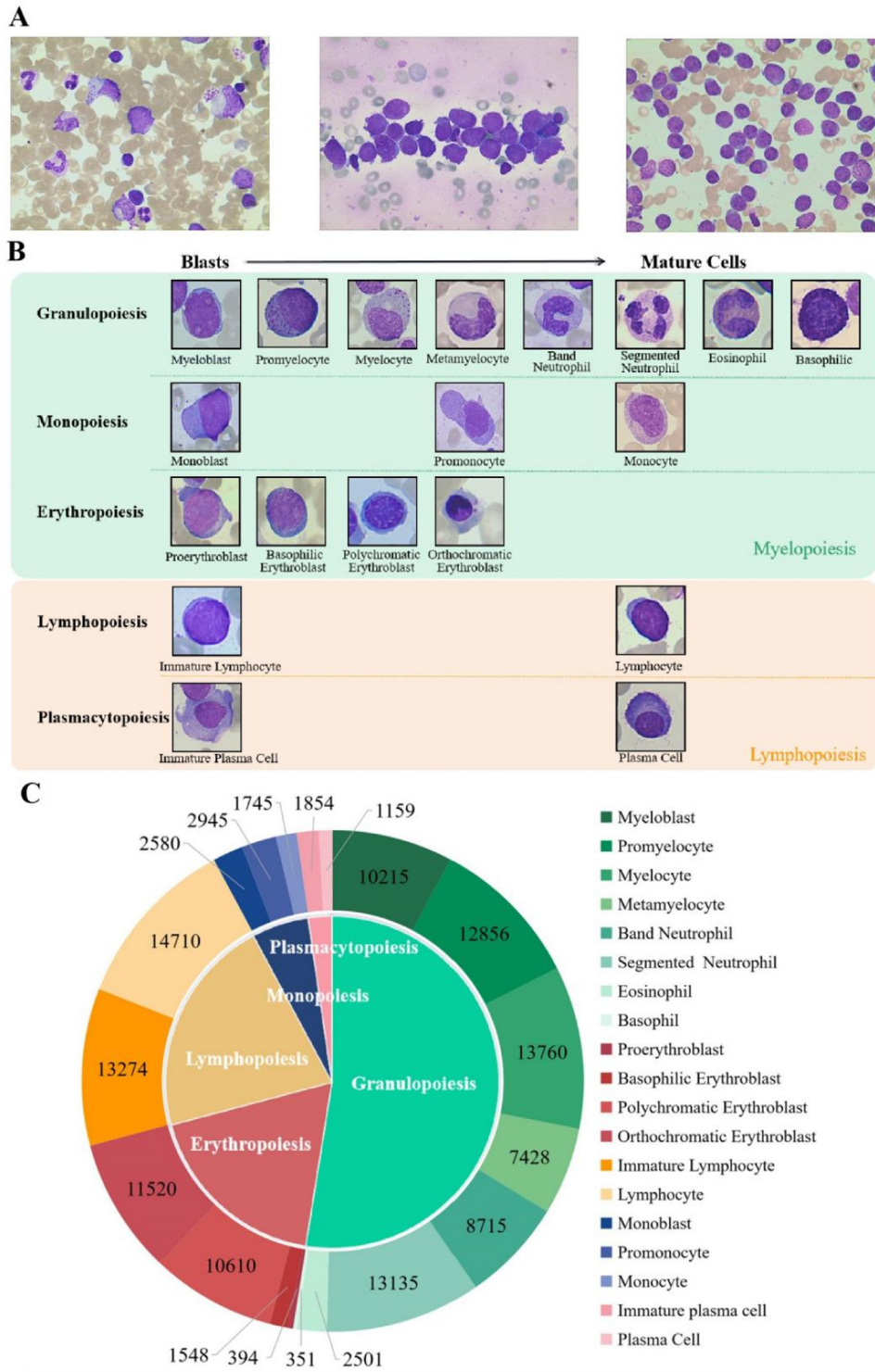


Fig. 2. Information of our data set. (A) Three examples of micrographs with different degrees of cell density. (B) The development and evolution of BM cells and 19 fine-grained categories highly correlated with the diagnosis. (C) The quantity distribution of the single cells. All smears were stained using the Wright Stain, and imaged at $\times 100$ magnification.

where α and β were both set to be 1 and y represented the actual label.

When predicting the label of a single cell, we preprocessed the image and sent it into MLFL-Net to acquire four different classification probabilities. We averaged these four probabilities and selected the category with a max probability as the final prediction.

2.4. Prediction of Leukemia Types

With the predicted labels of single cells, it is simple to automatically calculate key ratios of BM cells for each smear. According to the comparison of key ratios and the clinical standard, we can predict the leukemia types of patients. In our work, we took

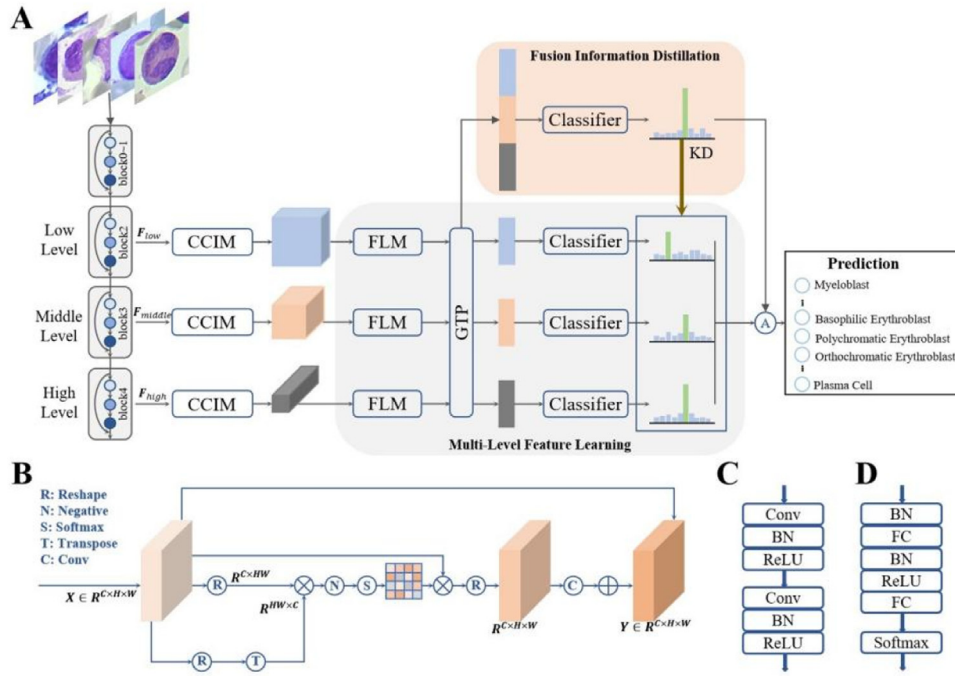


Fig. 3. The architecture of our model for the fine-grained classification. (A) The network architecture of MLFL-Net. (B) The process of CCIM module. (C) The architecture of FL module. (D) The architecture of the Classifier.

some common acute leukemia (AL) cases as an example for artificial intelligence-assisted diagnosis support system. The authoritative WHO standard [20] divides AL into different subtypes. There were four subtypes in our test cases including Acute lymphoblastic leukemia/lymphoma, Acute myeloid leukemia with maturation, Acute myelomonocytic leukemia and Acute monoblastic/monocytic leukemia, which have clear clinical decision points in hematology of WHO standard. Without doubt, the artificial intelligence-assisted diagnosis support system will greatly improve the efficiency and promote the development of medical science.

3. Evaluations and results

We conducted some experiments to evaluate the artificial intelligence-assisted diagnosis support system. First, we used the dataset of micrographs to train and evaluate a novel model YOLOX-s for cell detection. In the second part, we measured the performance of our proposed classification model MLFL-Net. Finally, we took some acute leukemia (AL) cases as an example to predict the leukemia types and evaluate the diagnosis support system. All experiments in the present work were carried out in the python environment using python 3.7, pytorch version 1.4.0 and Cuda version 10.0. The detection network was trained on NVIDIA Titan Xp graphics processing units and the classification network was trained on NVIDIA Geforce GTX 1080 graphics processing units.

3.1. Detection results

We evaluated the detection model in the validation set of 2358 micrographs and the results were shown in Fig. 4. The average inference time for an micrograph was 19.88ms and the model can accurately detect every single cell in micrographs. We took three micrographs as examples to show our detection performance. The three micrographs were from three different patients with different degrees of cell density, including different kinds of cells. As shown in Fig. 4, almost all the cells were exactly detected, which met the requirements of BM cell detection.

For a quantitative evaluation, we adopted precision-recall curve [30] and bounding box placement error [3] as assessment measures. The area under precision recall curve(AUC) is one of the most widely used metrics for model evaluation which measures detection accuracy over a broad range of detection sensitivities. It is shown in Fig. 5A that our detection model achieved high precision and recall with 0.9793 AUC. In addition, we utilized box placement error which is defined as the error in the positioning difference between the predicted box centers and the matched manually-annotated box centers. We calculated the Euclidean distance between box centers and normalized by the manually-annotated bounding box size (using half the length of the annotated box diagonal). Correct bounding box placement is critical for the classification stage, since the center of the predicted boxes is used to extract single cell images for classification. Shown in Fig. 5B, the median relative box placement error was only 4.33%, indicating good coincidence between the centers of predicted and actual cell centers.

3.2. Classification results

To prove the effectiveness of our proposed MLFL-Net, we evaluated it on the validation set of single cells. In order to directly present the performance, we showed the confusion matrix of MLFL-Net in Fig. 6. The deep color in the main diagonal here represented that most of the samples were correctly predicted, which obviously proved the effectiveness of our model.

For a quantitative evaluation, we adopted accuracy, precision and recall as assessment measures, defined as:

$$accuracy = \frac{\text{correct samples}}{\text{total samples}}$$

$$precision = \frac{\text{true positive}}{\text{true positive} + \text{false positive}}$$

$$recall = \frac{\text{true positive}}{\text{true positive} + \text{false negative}}$$

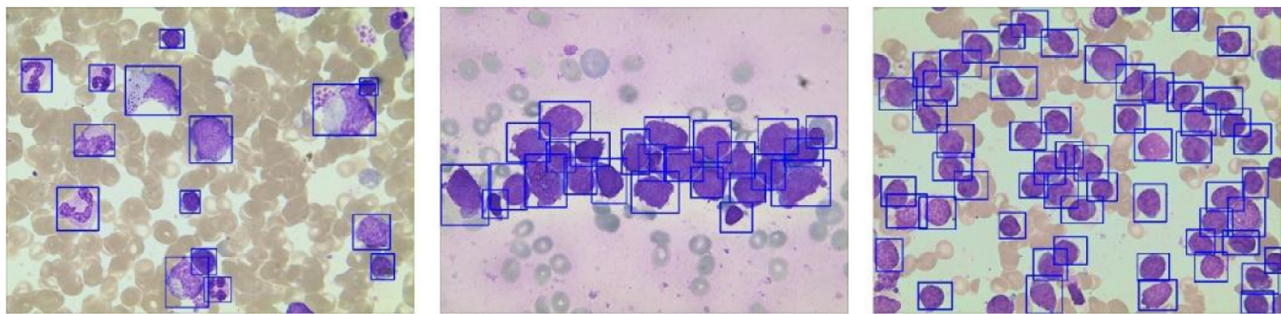


Fig. 4. Cell detection results. Three example micrographs and the detected bounding boxes in them. In micrographs with different degrees of cell density, the bounding boxes were highly corresponding to the ground truth.

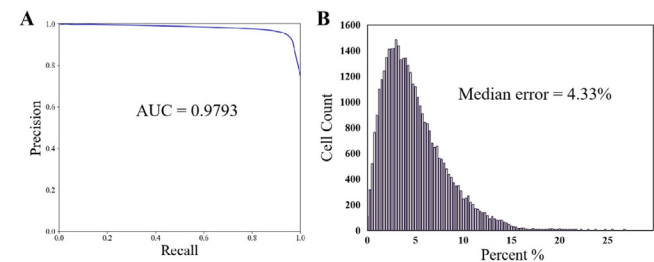


Fig. 5. Cell detection results. (A) Precision-recall of detection algorithm. (B) Histogram of bounding box placement error. This error measures the distance between predicted and actual bounding box centers relative to actual bounding box size.

where true positive and true negative are defined as the number of images that are classified or not classified, respectively, into a given class in agreement with the ground truth. Similarly, false positive

and false negative signify the number of images that are classified or not classified, respectively, into a given class in disagreement with the ground truth [17].

Concretely, we calculated the accuracy of our model and meanwhile the recall and precision for each category. The total accuracy of our model reached 89.53%. Table 1 showed the precision and the recall of each category respectively. Among the total 19 classes, 8 classes of cells had a precision above 90% and all classes of cells had a precision above 75%. As for the recall, 8 classes of cells had a recall above 90% and only the recall of Monoblast was below 70%.

In addition, we compared our proposed MLFL-Net with some other latest classification models. We conducted experiments on our data set using different models and the results were shown in Table 2. WBCNet [31] was a self-built model utilizing classical networks as reference for white blood cell classification and achieved accuracy of 85.49%, which was 4.04% below the accuracy of our model. MMAL-Net [32] was well designed with attention mechanism for fine-grained classification. It achieved a great performance

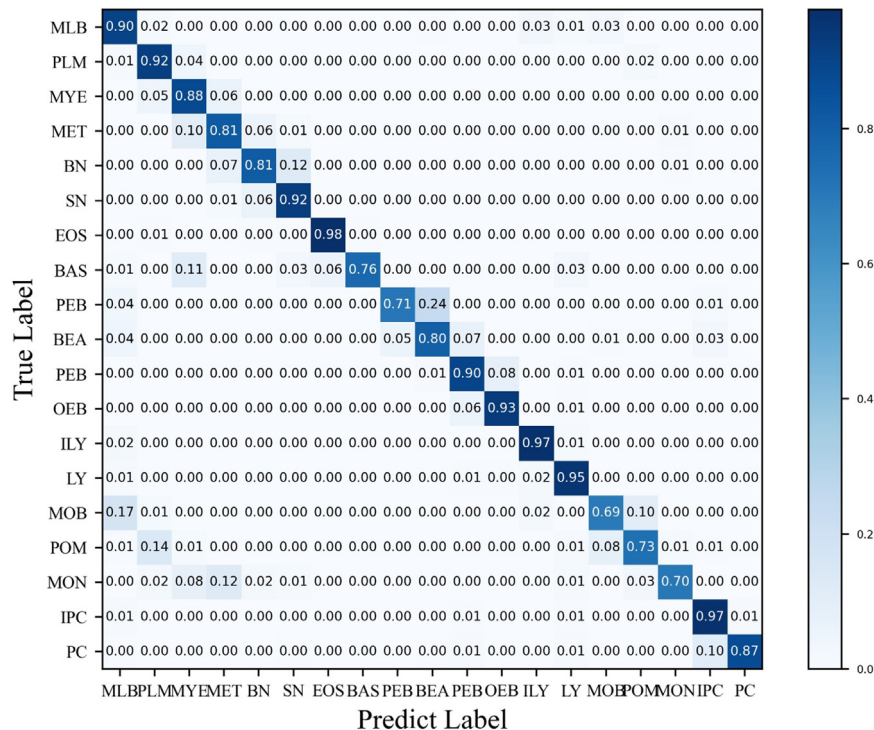


Fig. 6. The confusion matrix of our MLFL-Net model in our validation set. Deep blue color in the matrix diagonal entries shows the high accuracy. Abbreviations of cell names are used in the figure. (Abbreviations and corresponding names are listed here: MLB: Myeloblast, PLM: Promyelocyte, MYE: Myelocyte, MET: Metamyelocyte, BN: Band Neutrophil, SN: Segmented Neutrophil, EOS: Eosinophil, BAS: Basophil, PEB: Proerythroblast, BEA: Basophilic Erythroblast, PEB: Polychromatic Erythroblast, OEB: Orthochromatic Erythroblast, ILY: Immature Lymphocyte, LY: Lymphocyte, MOB: Monoblast, POM: Promonocyte, MON: Monocyte, IPC: Immature Plasma Cell, PC: Plasma Cell).

Table 1
Class-wise precision and recall of the classifier.

Class	Precision	Recall
Myeloblast	0.8930	0.9026
Promyelocyte	0.8955	0.9160
Myelocyte	0.8838	0.8761
Metamyelocyte	0.7644	0.8122
Band Neutrophil	0.8444	0.8067
Segmented Neutrophil	0.9101	0.9212
Eosinophil	0.9760	0.9760
Basophil	0.9310	0.7606
Proerythroblast	0.7671	0.7089
Basophilic Erythroblast	0.8179	0.7968
Polychromatic Erythroblast	0.9119	0.8973
Orthochromatic Erythroblast	0.9224	0.9340
Immature Lymphocyte	0.9487	0.9676
Lymphocyte	0.9580	0.9531
Monoblast	0.7558	0.6944
Promonocyte	0.7673	0.7334
Monocyte	0.8289	0.7037
Immature Plasma Cell	0.8908	0.9677
Plasma Cell	0.9526	0.8664

Table 2
Accuracy comparison with other methods.

Model	Accuracy (%)
WBCNet	85.49
MMAL-Net	84.35
ResNet-50	87.07
MLFL-Net (our proposed model)	89.53

in the general fine-grained image data set but gained only 84.35% accuracy on our data set, which was 5.18% lower than that of our model. The results of WBCNet and MMAL-Net indicated that directly applying existing models led to poor results for cell classification. Also, we compared our model with ResNet-50 [26] and the accuracy of our model was 2.46% higher than that of ResNet-50, which proved that the network designed for BM cells was better than classical CNN architectures. The results shown in Table 2 illustrated that our proposed method outperformed all the other methods.

Furthermore, to test the generalizability of our MLFL-Net, we evaluated our model on the data set of BM cells published by Matek et al. [17]. We randomly selected 80% of the available images in each class for training, whereas 20% were used for evaluation of the trained network. In a direct performance comparison of the sequential model, ResNeXt-50 utilized by Matek et al. [17] and our proposed MLFL-Net, MLFL-Net outperformed ResNeXt-50 in most of the classes as shown in Fig. 7. The recalls of MLFL-Net for most categories were much higher than ResNeXt-50. Specially, Blasts and Segmented neutrophils got highest improvement(23%). Overall, in the evaluation period, we gained the total accuracy of 89.09%. We calculated the total accuracy according to the precision and recall reported by Matek et al and gained the result of 72.28%, which was far below our result. It strongly proved the effectiveness of our proposed MLFL-Net.

At present, lots of researchers have been exploring the interpretability of CNN and Grad-CAM [33] is a famous visualization tool which is widely used. For an input image, it can generate a heatmap, in which the area with deeper color represents the area considered more important. In order to illustrate the important regions, we adopted it to visualize four cells of different kinds and the heatmaps were shown in Fig. 8. The first row showed the original image while the second, third and fourth row represented the low level, middle level and high level respectively. Through Fig. 8A, we found that three levels concentrated on different areas of cells

Table 3
The result of assisted diagnosis for acute leukemia.

Types	Total Number	Correct	False
Acute lymphoblastic leukemia/lymphoma	16	16	0
Acute myeloid leukemia with maturation	16	14	2
Acute myelomonocytic leukemia	5	1	4
Acute monoblastic/monocytic leukemia	3	3	0
Normal	40	40	0
Total	80	74	6

and gained information with different granularities, which proved that our MLFL-Net can focus on cell contour, cytoplasm and nucleus in low level, middle level and high level respectively. By concatenating three granularities of cells, the network obtained more comprehensive information of cells, which proved the effectiveness of our method to a certain extent.

Meanwhile, in order to directly show the effectiveness of our model, we extracted the feature vectors in the FID branch and reduced the dimensions into 2 from 3072 by using Uniform Manifold Approximation and Projection (UMAP) algorithm [34]. Fig. 8B showed the result of the projection and a single point represented a single cell. From the figure, we can easily find out that each class has its own feature space, which means the classifier can generally separate individual classes well due to the extracted features. Whereas, some classes were proximate to other classes in the feature embedding space, which illustrated a high degree of confusion between these classes.

3.3. Diseases prediction results

After the detection and classification stages, we obtained single cells and their categories in each micrograph, according to which we can easily gain the ratios of different kinds of cells for every patient. Thus, we can predict the leukemia types based on the proportion of different cells. We took some acute leukemia (AL) cases as an example to evaluate the diagnosis support system.

We utilized a data set including 40 acute leukemia (AL) patients and 40 normal people (BM transplantation donors) to evaluate the effectiveness of our method. We followed the diagnosis support procedure of cell detection, cell classification and prediction of leukemia types according to the WHO standard [21]. For example, when the proportion of Immature Lymphocytes is more than 20%, the computer will give a prediction result "Acute lymphoblastic leukemia". When the proportion of Monoblasts and Promonocytes is more than 20%, the computer will give a prediction result "Acute monoblastic/monocytic leukemia".

As shown in Table 3, forty normal people were totally correctly predicted, proving the great performance in distinguish normal people and patients of our procedure. For the diagnosis of different AL patients, some disease achieved 100% accuracy, including Acute lymphoblastic leukemia/lymphoma and Acute monoblastic/monocytic leukemia. Meanwhile, some patients with Acute myelomonocytic leukemia were predicted as Acute myeloid leukemia with maturation. This is due to lower monoblasts classification performance, where many monoblasts were classified into myeloblasts, similar to expert human hematopathologists. In general, the total accuracy was 92.5%, reflecting the potential for assisted diagnosis.

Overall, in the artificial intelligence-assisted diagnosis support system, the novel YOLOX-s model detected cells precisely and obtained single cell images. Our proposed MLFL-Net obtained the total accuracy 89.53% and outperformed other models. Furthermore, 92.5% of the prediction results for the cohort were consistent with the ones gave by experts. Our study is of great significance for the assisted diagnosis of hematologic diseases.

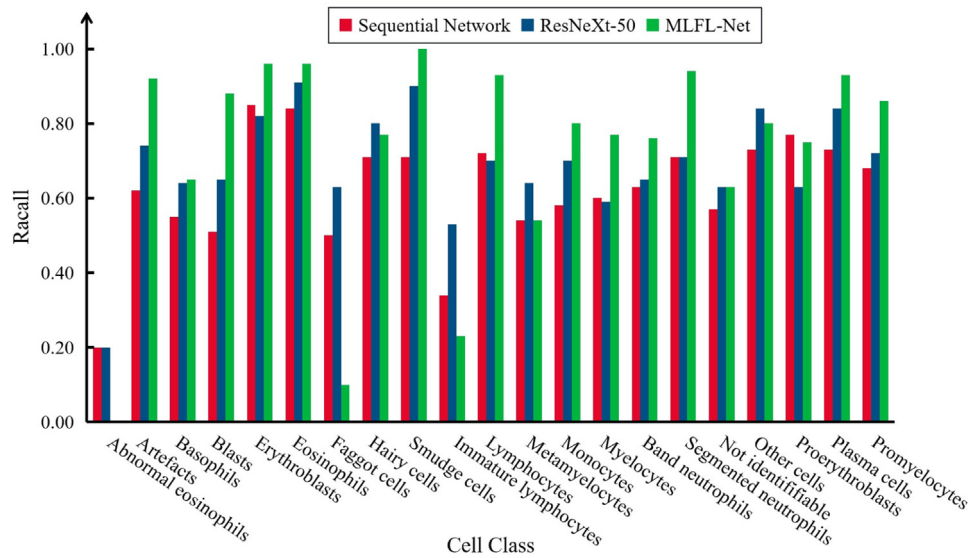


Fig. 7. The comparison of recall for each class in three different models on the data set of BM cells published by Matek et al.

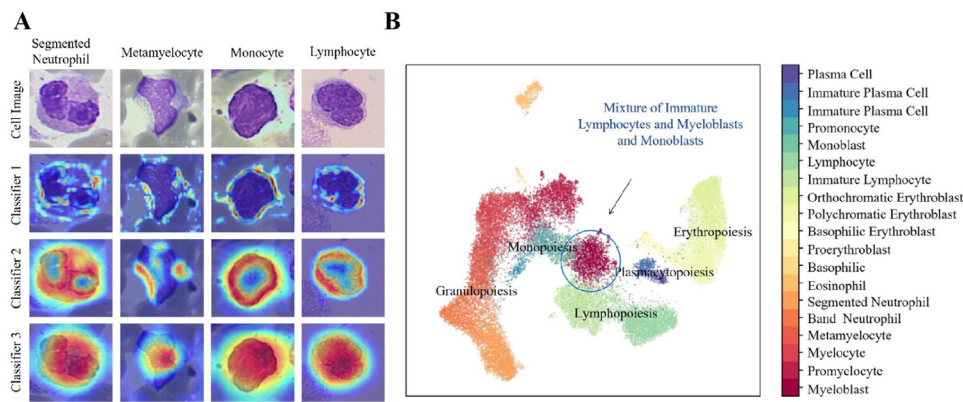


Fig. 8. The visualization of the features. (A) Four cells of different kinds and the heatmaps of features at different levels. (B) The UMAP embedding of extracted features in the FID branch.

4. Discussion

In this paper, we present an artificial intelligence-assisted diagnosis support system of morphological examination based on bone marrow smears including detection, classification and prediction of leukemia types. This system performed well, with high precision and accuracy in both detection and cell classification. Meanwhile, we constructed a large-scale well-annotated data set of BM cells, consisting of 11,788 micrographs and 131,300 single cell images, which was the strong foundation of our study.

We followed the procedure of detection first and classification later. We used the data set of micrographs to train and evaluate a novel model YOLOX-s for cell detection. The results were remarkable with high accuracy in spite that images were equipped with different degrees of cell density, which was the basis for cell classification. The classification of BM cells is one of the most important steps in the procedure of diagnosis. We regarded it as a fine-grained classification task and proposed a novel architecture called MLFL-Net, which was the first one to adopt multi-level features for BM cell classification as we know. The results in our data set were encouraging with a high accuracy, outperforming the baseline model and other related models. Meanwhile, the recall and precision values were high enough to meet the demand for assisted diagnosis. After detection and classification stages, every single cell in micrographs was classified to a specific category. Then the key

ratios of different cell categories for every BM smear was calculated, which were important evidences for disease prediction. We evaluated the whole diagnosis support procedure on 40 slides from different kinds of AL patients and 40 slides from donors. According to the WHO standard for prediction of AL, our method gained a total accuracy of 92.5%, which showed the potential for assisted diagnosis.

At present, our system can provide diagnosis support for some types of leukemia when bone marrow micrographs of the patient are available. However, there is still room for improvement of our system. For example, the imbalance problem in classification is a noted problem in AI-based bone marrow studies. Some classes, like Basophil and Proerythroblast, got relatively low accuracy. This is likely due to the rarity of these objects, and performance may improve with access to large training datasets. In addition, due to having similar structure with granulocytes, many monocytes including Monoblast, Promonocyte and Monocyte were misclassified into granulocytes, which lead to relatively low recall in these classes, similar to expert human hematopathologists. In the future, we will try more methods to improve classification accuracy, including concentrating more on cell classes that are easily confused, and complementing the data including various diseases and the abnormal cells to make it more universal.

In the meantime, the adaptability of the system needs to be improved. There are big differences in the bone marrow images gath-

ered from different devices with different resolution and shade of staining. Therefore, to keep the accurate classification result, the model should be retrained with new data before it is ready for a certain hospital. It is laborious and unpractical to collect and label data for each hospital. Therefore, we will introduce mechanism of the normalization of staining and domain adaptation methods to improve the robustness of our method in the future. As well, another limitation is that our diagnosis support system can only provide morphological information for doctors. In clinical practice, doctors combine the results of multiple examinations to give diagnosis. Furthermore, we anticipate including data from flow cytometry or molecular genetics in our methods to construct a multi-modal deep learning network and improve the integrity of the procedure.

Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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